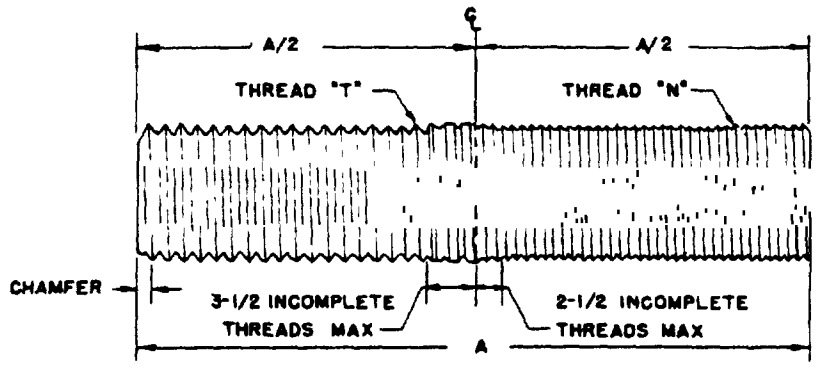
FED SUP CLASS
5307



REVIEW ACTIVITIES: AV, IS, NSA, 85
USER ACTIVITIES: AT, CE, EL, MC, ME, MI, OS, SH

TABLE I - (TAP END THREAD DESIGNATION CLASS 5HF)

DASH NO (STEEL)	T TAP END THREAD NC-5 HF	N NUT END THREAD UNF-2A
102	2500-20	2500-28
103	3125-18	3125-24
104	.3750-16	3750-24
105	4375-14	4375-20
106	.5000-13	5000-20
107	5625-12	5625-18
108	6250-11	6250-18
109	7500-10	7500-16
110	8750- 9	8750-14
111	1.0000- 8	1 0000-12
112	1 1250- 7	1.1250-12
113	1 2500- 7	1.2500-12

TABLE II - (TAP END THREAD DESIGNATION CLASS 5 CSF)

DASH NO (CRES)	DASH NO (BRASS)	T TAP END THREAD NC-5 CSF	N NUT END THREAD UNF-2A
202	302	2500-20	2500-28
203	303	3125-18	3125-24
204	304	3750-16	3750-24
205	305	4375-14	4375-20
206	306	5000-13	5000-20
207	307	5625-12	5625-18
208	308	6250-11	6250-18
209	309	7500-10	7500-16
210	310	.8750- 9	8750-14
211	311	1 0000- 8	1 0000-12
212	312	1 1250- 7	1 1250-12
213	313	1 2500- 7	1 2500-12

TABLE IIB - (TAP END THREAD DESIGNATION CLASS 5 ONF)

DASH NO (NI-CU)	T TAP END THREAD NC-5 ONF	N NUT END THREAD UNF-2A
402	2500-20	2500-28
404	3750-16	3750-24
406	5000-13	5000-20
408	6250-11	6250-18
409	7500-10	7500-16
410	8750- 9	8750-14
411	1 0000- 8	1 0000-12
412	1 1250- 7	1 1250-12
413	1 2500- 7	1 2500-12

NOTES

1. MATERIAL: Steel, carbon or alloy, per Federal Standard No 66
   Steel, corrosion resisting, per Federal Standard No 66, 300 Series.
   Brass, per QQ-B-637, Composition 2
   Nickel-copper alloy, per QQ-N-281, Class B, cold drawn or equivalent
2. PROTECTIVE COATING: Steel, carbon or alloy, cadmium plated per QQ-P-416, Type II, Class 3
   Steel, corrosion resisting, passivated per QQ-P-35
   Brass, oxide or black finish per MIL-F-495
3. THREADS: NC-5 threads shall conform to NBS Handbook H28 Part III
   UNF-2A threads shall conform to NBS Handbook H28 Part I
4. MECHANICAL PROPERTIES: Studs shall be capable of developing mechanical properties in accordance with the procurement specification. Tensile strengths and pertinent length dash numbers are listed below

Material	Min Tensile Strength PSI	Dash Numbers
Steel carbon or alloy		
Sizes 1/4 thru 1	120,000	
Over 1 to 1-1/2	105 000	102-113
Steel, cres	80 000	202-213
Brass	60,000	302-313
Nickel-copper alloy	80,000	402-413

5. HARDNESS: Steel, carbon or alloy, Rockwell C19 minimum
   Steel, cres, Brinell 140 minimum.
   Brass, Rockwell B65 minimum
   Nickel-copper alloy, Rockwell B80 minimum
6. CHAMFER: 45° to 50° angle from the outside diameter to 1/64 to 1/32 inch below root diameter of thread
7. DIMENSIONS: Dimensions in inches, to be met after plating
8. LENGTH TOLERANCES: Studs 6 inches and under ± 125 inch studs over 6 inches ± 188 inch
9. PART NUMBERS: The MS part number consists of the MS number, plus the dash number, plus the second dash number for length (table IV). Example
   MS51864-102-16 stud, carbon or alloy steel, 2500 inch dia , 2 inch length.
   MS51864-204-18 stud, cres, 3750 inch dia , 2.250 inch length
   MS51864-306-20 stud, brass, 5000 inch dia , 2 500 inch length
   MS51864-409-30 stud, nickel-copper alloy, .7500 inch dia , 3 750 inch length
10. For design feature purposes, this standard takes preference over procurement documents referenced herein.
11. Referenced documents shall be of the issue in effect on date of invitations for bid
12. Illustration is for identification and not intended to restrict design and shapes of studs which otherwise conform to requirements

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P A MC Other Cust MZ	TITLE: STUD- [illegible]AIN, GENERAL PURPOSE	MILITARY STANDARD MS51864
PROCUREMENT SPECIFICATION FF-S-1362	SUPERSEDES MS15986, MS15987, MS15988 MS15989, MS15990 MS15991	SHEET 1 OF 4

DD FORM 672-1 (Continued)

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TABLE IV

LENGTH DASH NUMBER	DIAMETER DASH NUMBERS											
	102 202 302 402		103 203 303		104 204 304 404		105 205 305		106 206 306 406		107 207 307	
	A	A/2	A	A/2	A	A/2	A	A/2	A	A/2	A	A/2
12	1 500	750										
14	1 750	875	1.750	875								
16	2 000	1 000	2 000	1 000	2 000	1 000						
18	2 250	1 125	2 250	1 125	2 250	1 125	2 250	1 125				
20	2 500	1 250	2 500	1 250	2 500	1 250	2 500	1.250	2 500	1.250		
22	2 750	1 375	2 750	1 375	2 750	1 375	2 750	1 375	2 750	1 375	2 750	1 375
24	3 000	1 500	3 000	1 500	3 000	1 500	3 000	1.500	3 000	1 500	3 000	1 500
26					3 250	1 625	3 250	1 625	3 250	1 625	3 250	1 625
28					3 500	1 750	3 500	1 750	3 500	1 750	3 500	1 750
30					3 750	1 875	3 750	1 875	3 750	1 875	3 750	1 875
32					4 000	2 000	4 000	2.000	4.000	2 000	4.000	2 000
34									4.250	2 125	4 250	2 125
36									4 500	2 250	4 500	2 250
38									4 750	2 375	4 750	2 375
40									5 000	2 500	5.000	2.500
44									5 500	2 750	5 500	2 750
48									6 000	3 000	6 000	3.000

LENGTH DASH NUMBER	DIAMETER DASH NUMBERS											
	108 208 308 408		109 209 309 409		110 210 310 410		111 211 311 411		112 212 312 412		213 213 313 413	
	A	A/2	A	A/2	A	A/2	A	A/2	A	A/2	A	A/2
24	3 000	1 500										
26	3.250	1 625										
28	3 500	1 750	3 500	1 750								
30	3 750	1 875	3.750	1 875								
32	4 000	2.000	4 000	2 000	4 000	2 000						
34	4.250	2 125	4 250	2 125	4 250	2.125						
36	4 500	2 250	4 500	2 250	4 500	2.250	4 500	2 250				
38	4 750	2 375	4 750	2 375	4 750	2 375	4 750	2 375	4 750	2.375		
40	5 000	2 500	5 000	2 500	5 000	2 500	5 000	2 500	5 000	2 500		
44	5 500	2 750	5 500	2 750	5 500	2 750	5 500	2 750	5 500	2 750	5 500	2 750
48	6 000	3 000	6 000	3 000	6 000	3.000	6 000	3 000	6 000	3 000	6 000	3 000
52									6 500	3.250	6 500	3 250
56									7 000	3 500	7 000	3 500
60									7 500	3 750	7 500	3 750
64									8 000	4 000	8 000	4 000

REVIEW: AV, IS, NSA, 85
USER: AT, CE, EL, MC, ME, MI, OS, SH

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P.A. MC
Other Coast 82

TITLE: STUD — PLAIN, GENERAL PURPOSE

MILITARY STANDARD
MS51864

PROCUREMENT SPECIFICATION: FF-S-1362

SUPERSEDES: MS15986, MS15987, MS15988, MS15989, MS15990, MS15991

SHEET 2 OF

DD FORM 672-1 (Reverse)

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REVIEW AY, IS, NSA, 85
USER AT, CE, EL, MC, ME, MI, OS, SH

TABLE V - INTERCHANGEABILITY

The studs covered by dash numbers given in the original issues of MS15986, MS15987, MS15988, MS15990, MS15991, and Revision A of MS15989, are cancelled after the original date of this document. Use the dash numbers given in the preceding sheets. The cancelled studs cannot always replace the new studs and should be used until existing stocks are depleted. Use only the new studs for new design and replacement. Replacement shall be in accordance with this table.

CROSS REFERENCE

CANCELLED			NEW	CANCELLED			NEW	CANCELLED			NEW
MS15986		MS15986 MS15989	MS51864	MS15986		MS15986 MS15989	MS51864	MS15986		MS15986 MS15989	MS51864
F11*	FC11*	FP11*	102-12	J21*	JC21*	JP21*	105-22	N34	NC34	NP34	108-34
F12	FC12	FP12	102-12	J22	JC22	JP22	105-22	N36	NC36	NP36	108-36
F13*	FC13*	FP13*	102-14	J23*	JC23*	JP23*	105-24	N38	NC38	NP38	108-38
F14	FC14	FP14	102-14	J24	JC24	JP24	105-24	N40	NC40	NP40	108-40
F15*	FC15*	FP15*	102-16	J26	JC26	JP26	105-26	N44	NC44	NP44	108-44
F16	FC16	FP16	102-16	J28	JC28	JP28	105-28	N48	NC48	NP48	108-48
F17*	FC17*	FP17*	102-18	J30	JC30	JP30	105-30				
F18	FC18	FP18	102-18	J32	JC32	JP32	105-32				
F19*	FC19*	FP19*	102-20					P28	PC28	PP28	109-28
F20	FC20	FP20	102-20					P30	PC30	PP30	109-30
F21*	FC21*	FP21*	102-22	K19*	KC19*	KP19*	106-20	P32	PC32	PP32	109-32
F22	FC22	FP22	102-22	K20	KC20	KP20	106-20	P34	PC34	PP34	109-34
F23*	FC23*	FP23*	102-24	K21*	KC21*	KP21*	106-22	P36	PC36	PP36	109-36
F24	FC24	FP24	102-24	K22	KC22	KP22	106-22	P38	PC38	PP38	109-38
				K23*	KC23*	KP23*	106-24	P40	PC40	PP40	109-40
				K24	KC24	KP24	106-24	P44	PC44	PP44	109-44
G13*	GC13*	GP13*	103-14	K26	KC26	KP26	106-26	P48	PC48	PP48	109-48
G14	GC14	GP14	103-14	K28	KC28	KP28	106-28				
G15*	GC15*	GP15*	103-16	K30	KC30	KP30	106-30				
G16	GC16	GP16	103-16	K32	KC32	KP32	106-32	Q32	QC32	QP32	110-32
G17*	GC17*	GP17*	103-18	K34	KC34	KP34	106-34	Q34	QC34	QP34	110-34
G18	GC18	GP18	103-18	K36	KC36	KP36	106-36	Q36	QC36	QP36	110-36
G19*	GC19*	GP19*	103-20	K38	KC38	KP38	106-38	Q38	QC38	QP38	110-38
G20	GC20	GP20	103-20	K40	KC40	KP40	106-40	Q40	QC40	QP40	110-40
G21*	GC21*	GP21*	103-22	K44	KC44	KP44	106-44	Q44	QC44	QP44	110-44
G22	GC22	GP22	103-22	K48	KC48	KP48	106-48	Q48	QC48	QP48	110-48
G23*	GC23*	GP23*	103-24								
G24	GC24	GP24	103-24								
				M21*	MC21*	MP21*	107-22	R36	RC36	RP36	111-36
				M22	MC22	MP22	107-22	R38	RC38	RP38	111-38
H15*	HC15*	HP15*	104-16	M23*	MC23*	MP23*	107-24	R40	RC40	RP40	111-40
H16	HC16	HP16	104-16	M24	MC24	MP24	107-24	R44	RC44	RP44	111-44
H17*	HC17*	HP17*	104-18	M26	MC26	MP26	107-26	R48	RC48	RP48	111-48
H18	HC18	HP18	104-18	M28	MC28	MP28	107-28				
H19*	HC19*	HP19*	104-20	M30	MC30	MP30	107-30				
H20	HC20	HP20	104-20	M32	MC32	MP32	107-32	S38	SC38	SP38	112-38
H21*	HC21*	HP21*	104-22	M34	MC34	MP34	107-34	S40	SC40	SP40	112-40
H22	HC22	HP22	104-22	M36	MC36	MP36	107-36	S44	SC44	SP44	112-44
H23*	HC23*	HP23*	104-24	M38	MC38	MP38	107-38	S48	SC48	SP48	112-48
H24	HC24	HP24	104-24	M40	MC40	MP40	107-40	S52	SC52	SP52	112-52
H26	HC26	HP26	104-26	M44	MC44	MP44	107-44	S56	SC56	SP56	112-56
H28	HC28	HP28	104-28	M48	MC48	MP48	107-48	S60	SC60	SP60	112-60
H30	HC30	HP30	104-30					S64	SC64	SP64	112-64
H32	HC32	HP32	104-32								
				N23*	NC23*	NP23*	108-24	T44	TC44	TP44	113-44
				N24	NC24	NP24	108-24	T48	TC48	TP48	113-48
J17*	JC17*	JP17*	105-18	N26	NC26	NP26	108-26	T52	TC52	TP52	113-52
J18	JC18	JP18	105-18	N28	NC28	NP28	108-28	T56	TC56	TP56	113-56
J19*	JC19*	JP19*	105-20	N30	NC30	NP30	108-30	T60	TC60	TP60	113-60
J20	JC20	JP20	105-20	N32	NC32	NP32	108-32	T64	TC64	TP64	113-64

* Next higher increment is indicated and recommended for replacement

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P.A. MC Other Cost 82	TITLE STUD — PLAIN, GENERAL PURPOSE	MILITARY STANDARD MS51864
PROCUREMENT SPECIFICATION FF-S-1362	SUPERSEDES MS15986, MS15987, MS15988, MS15989, MS15990, MS15991.	SHEET 3 OF

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REVIEWER: AV, IS, NSA, RS
USAF AT, CR, EL, MC, ME, MI, CS, SH

TABLE V - INTERCHANGEABILITY (CONTINUED)

CROSS REFERENCE			
CANCELLED	NEW	CANCELLED	NEW
MS15987 MS15990	MS51864	MS15987 MS15990	MS51864
F12	302-12	M38	307-38
F14	302-14	M40	307-40
F16	302-16	M44	307-44
F18	302-18	M48	307-48
F20	302-20		
F22	302-22	N24	308-24
F24	302-24	N26	308-26
		N28	308-28
G14	303-14	N30	308-30
G16	303-16	N32	308-32
G18	303-18	N34	308-34
G20	303-20	N36	308-36
G22	303-22	N38	308-38
G24	303-24	N40	308-40
		N44	308-44
H16	304-16	N48	308-48
H18	304-18		
H20	304-20	P28	309-28
H22	304-22	P30	309-30
H24	304-24	P32	309-32
H26	304-26	P34	309-34
H28	304-28	P36	309-36
H30	304-30	P38	309-38
H32	304-32	P40	309-40
		P44	309-44
J18	305-18	P48	309-48
J20	305-20		
J22	305-22	Q32	310-32
J24	305-24	Q34	310-34
J26	305-26	Q36	310-36
J28	305-28	Q38	310-38
J30	305-30	Q40	310-40
J32	305-32	Q44	310-44
		Q48	310-48
K20	306-20		
K22	306-22	R36	311-36
K24	306-24	R38	311-38
K26	306-26	R40	311-40
K28	306-28	R44	311-44
K30	306-30	R48	311-48
K32	306-32		
K34	306-34	S38	312-38
K36	306-36	S40	312-40
K38	306-38	S44	312-44
K40	306-40	S48	312-48
K44	306-44	S52	312-52
K48	306-48	S56	312-56
		S60	312-60
M22	307-22	S64	312-64
M24	307-24		
M26	307-26	T44	313-44
M28	307-28	T48	313-48
M30	307-30	T52	313-52
M32	307-32	T56	313-56
M34	307-34	T60	313-60
M36	307-36	T64	313-64

CROSS REFERENCE					
CANCELLED		NEW	CANCELLED		NEW
MS15988	MS15991	MS51864	MS15988	MS15991	MS51864
F12	F12	402-12	M48	N48	408-48
F14	F14	402-14			
F16	F16	402-16			
F18	F18	402-18			
F20	F20	402-20			
F22	F22	402-22	N28	P28	409-28
F24	F24	402-24	N30	P30	409-30
			N32	P32	409-32
			N34	P34	409-34
			N36	P36	409-36
			N38	P38	409-38
			N40	P40	409-40
			N44	P44	409-44
H16	H16	404-16	N48	P48	409-48
H18	H18	404-18			
H20	H20	404-20			
H22	H22	404-22			
H24	H24	404-24			
H26	H26	404-26	P32	Q32	410-32
H28	H28	404-28	P34	Q34	410-34
H30	H30	404-30	P36	Q36	410-36
H32	H32	404-32	P38	Q38	410-38
			P40	Q40	410-40
			P44	Q44	410-44
			P48	Q48	410-48
J20	K20	406-20			
J22	K22	406-22	Q36	R36	411-36
J24	K24	406-24	Q38	R38	411-38
J26	K26	406-26	Q40	R40	411-40
J28	K28	406-28	Q44	R44	411-44
J30	K30	406-30	Q48	R48	411-48
J32	K32	406-32			
J34	K34	406-34			
J36	K36	406-36			
J38	K38	406-38			
J40	K40	406-40	R38	S38	412-38
J44	K44	406-44	R40	S40	412-40
J48	K48	406-48	R44	S44	412-44
			R48	S48	412-48
			R52	S52	412-52
			R56	S56	412-56
			R60	S60	412-60
			R64	S64	412-64
M24	N24	408-24			
M26	N26	408-26			
M28	N28	408-28			
M30	N30	408-30			
M32	N32	408-32	S44	T44	413-44
M34	N34	408-34	S48	T48	413-48
M36	N36	408-36	S52	T52	413-52
M38	N38	408-38	S56	T56	413-56
M40	N40	408-40	S60	T60	413-60
M44	N44	408-44	S64	T64	413-64

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P.A. WC	TITLE	MILITARY STANDARD
Other Cost 82	STUD — PLAIN, GENERAL PURPOSE	MS51864
PROCUREMENT SPECIFICATION FF-S-1362	SUPERSEDES MS15986 MS15987, MS15988, MS15989, MS15990, MS15991	SHEET 4 OF

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